

1. SMART Problem Statement

AstraPay is experiencing a divergence between top-line payment volume growth and bottom-line contribution profitability. Over the 5-day diagnostic period, the objective is to isolate the structural root causes - separating true business deterioration from data quality artifacts - and deliver a localized, data-backed 60-day action plan to stabilize and improve the contribution profit per successful transaction across the India, Singapore, and UAE markets.

2. KPIs, Scope, and Enabled Decisions

Executive KPI (The "North Star"):

- **Contribution Profit per Successful Transaction:**
 - *Numerator:* Gross Revenue (Merchant Pricing) – Processing Costs (Fixed + Variable) – Fraud/Chargeback Losses ± FX Spread Impact.
 - *Denominator:* Total count of successful, settled transactions.

Supporting KPIs:

- **Payment Success Rate:** (Successful Attempts / Total Attempts)
- **Average Processing Cost per Route:** Total route fees / Transactions processed via that route.
- **Fraud Decline Rate:** Transactions declined by risk rules / Total attempts.

Scope & Exclusions:

- **In Scope:** Retail and SME merchant transactions across IN, SG, and AE.
- **Exclusions:** Unsettled transactions (in-flight), duplicated records due to one-to-many operational joins, and artifacts caused by known system defects (e.g., missing identifiers or zero/negative amounts unless explicitly classified as valid refunds).
- **Assumptions:** The *transaction.csv* dataset is the anchor source of truth for payment attempts. If operational systems (like event logs) disagree, the transaction table takes precedence for volume metrics.

Decisions this Analysis Must Enable:

- Whether to re-negotiate pricing tiers for specific merchant segments.
- Whether to adjust routing logic to penalize high-latency/high-cost providers.
- Whether to recalibrate fraud rules to reduce false declines.

3. MECE Hypotheses & Evidence Criteria

To ensure our root-cause investigation is Mutually Exclusive and Collectively Exhaustive (MECE), we will test the following four hypotheses:

- **Hypothesis 1 (Commercial Mix-Shift):** Profitability per transaction is falling because volume growth is disproportionately concentrated in lower-margin merchant segments or unfavorable pricing plans.

- *Evidence to Confirm:* Segment analysis shows volume surging in tiers where fixed/variable revenue does not cover baseline processing costs.
- **Hypothesis 2 (Operational / Routing Drag):** Surges in processing latency are forcing transactions onto more expensive backup routes, inflating fixed and variable processing costs.
 - *Evidence to Confirm:* Cost per successful transaction spikes during specific periods, perfectly correlated with processing latency spikes on specific regional routes.
- **Hypothesis 3 (Risk / False Declines):** Overly aggressive fraud rules are declining legitimate volume, shrinking the denominator (successful transactions) while fixed operational costs remain constant, thereby reducing unit profitability.
 - *Evidence to Confirm:* A high volume of transactions with low-risk merchant profiles are triggering multiple fraud rules and being declined.
- **Hypothesis 4 (Data / FX Misalignment):** The perceived drop in profitability is a reporting illusion caused by the application of the wrong foreign exchange rate type (e.g., using Accounting rates instead of Settlement rates) or temporal leakage in data ingestion.
 - *Evidence to Confirm:* Recalculating the unit economics strictly using the correct *SETTLEMENT* FX rate eliminates the negative margin trend.